

Human-Computer Interaction A2 (2025/2026)

Team Name

Team Members: Guo jiahao, Kalile Horri, Intisar Bel meddeb

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1 Introduction

1.1 Team Members

- Jiahao Guo jiahao.guo@studio.unibo.it
- Kalike Horri kalile.horri@studio.unibo.it
- Intisar Bel meddeb Intisar.belmeddeb@studio.unibo.it

1.2 Solution overview

The proposed solution is a **Smart Sustainable Travel Companion** that guides users to make eco-friendly choices while exploring safely and enjoying authentic local experiences. It addresses the needs to *make sustainable choices easy, feel safe and in control, trust the system*, and *connect with local culture* by providing clear environmental indicators, safety information, transparent data usage, and motivational feedback. Users can share their experiences through a **recommendation system**, receive **friendly encouragement and reflection prompts**, complete **optional playful challenges**, and see the **positive impact of each small action**, celebrating everyday sustainable choices such as refilling bottles or walking instead of using transport. This solution was selected because it integrates planning, exploration, cultural engagement, and reflection into a seamless experience, supporting sustainable behaviors without extra effort or stress.

2 Task and storyboard

2.1 List of the simple, moderate, and complex tasks

- **1. Simple Task: Discover Eco-Friendly Options**
The user wants to quickly see sustainable travel choices.
- **2. Moderate Task: Plan a Safe and Enjoyable Trip**
The user wants to organize their day or trip, balancing sustainability, budget, and personal preferences.
- **3. Complex Task: Track, Reflect, and Share Sustainable Actions**
The user wants to monitor their sustainable choices, see their impact, complete optional challenges, and share experiences with others.

2.2 Storyboard



Figure 1: Storyboard illustrating the ecoGo user journey.

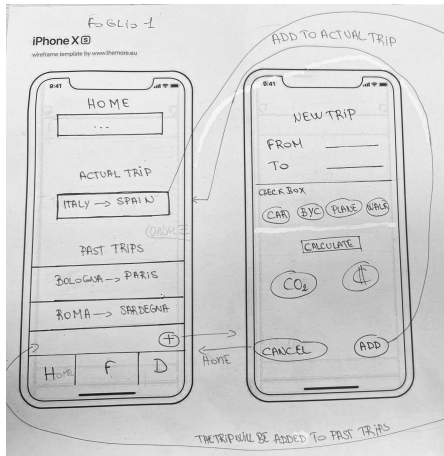
3 Modality exploration

3.1 Overview of the alternatives

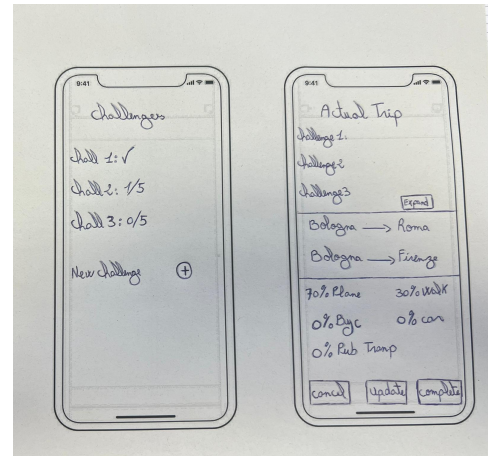
Modality 1: Web Application

A web app acts as the main platform where users can plan trips, compare eco-friendly travel bundles, and receive clear information about prices, safety, and environmental impact. It allows travelers to:

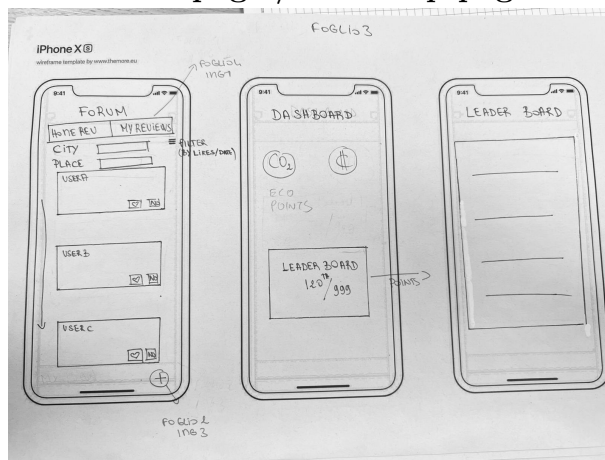
- View integrated transport options (metro, bus, electric vehicles).
- Receive simple, non-competitive feedback on sustainable choices.
- Access local eco-initiatives or events nearby.



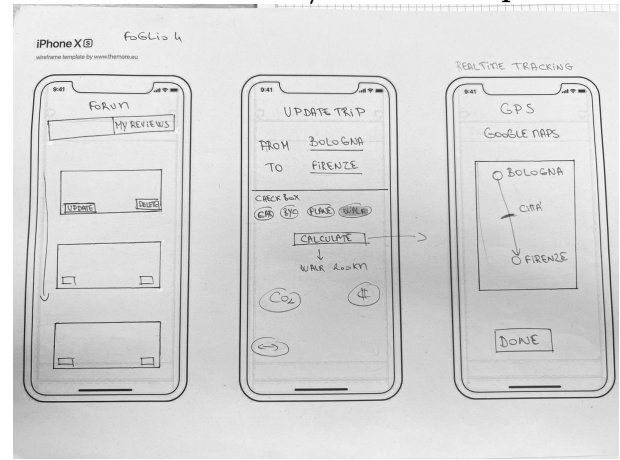
Home page / New trip page



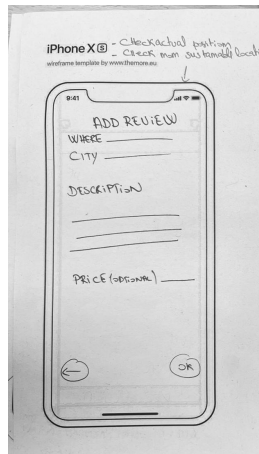
Gamification / Actual Trip



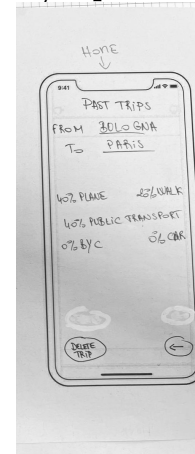
Eco-Score Dashboard



Forum reviews/ updating Trip/ Gps



Add review



Past Trips

Table 1: Real example illustrating the EcoGo user journey with web application.

This modality was chosen for its accessibility, clarity, and ability to centralize travel, cost, and sustainability information in one place.

Modality 2: Smart Glasses

The EcoGo Smart Glasses provide an immersive, hands-free experience that enhances sustainable travel through augmented reality (AR) and voice interaction. They offer:

- **Voice-controlled interface** for seamless, hands-free operation.
- **AR overlays** displaying eco-friendly routes, local green spots, and live navigation cues.
- **Real-time eco-suggestions** and sustainability tips triggered by context-aware sensors.
- **Eco-score visualization** showing personal CO₂ savings and progress toward eco-goals.
- **Gamification elements** such as badges, points, and achievements displayed in the heads-up view.
- **Instant sharing** of photos or reviews through voice command, integrating directly with the EcoGo app.

This modality was chosen for its ability to merge convenience, awareness, and motivation into one wearable solution. By blending AR guidance with real-time feedback, EcoGo Smart Glasses empower users to make sustainable choices naturally—without interrupting their travel flow. The lightweight design ensures comfort, while voice control and visual cues make eco-friendly living effortless and engaging.

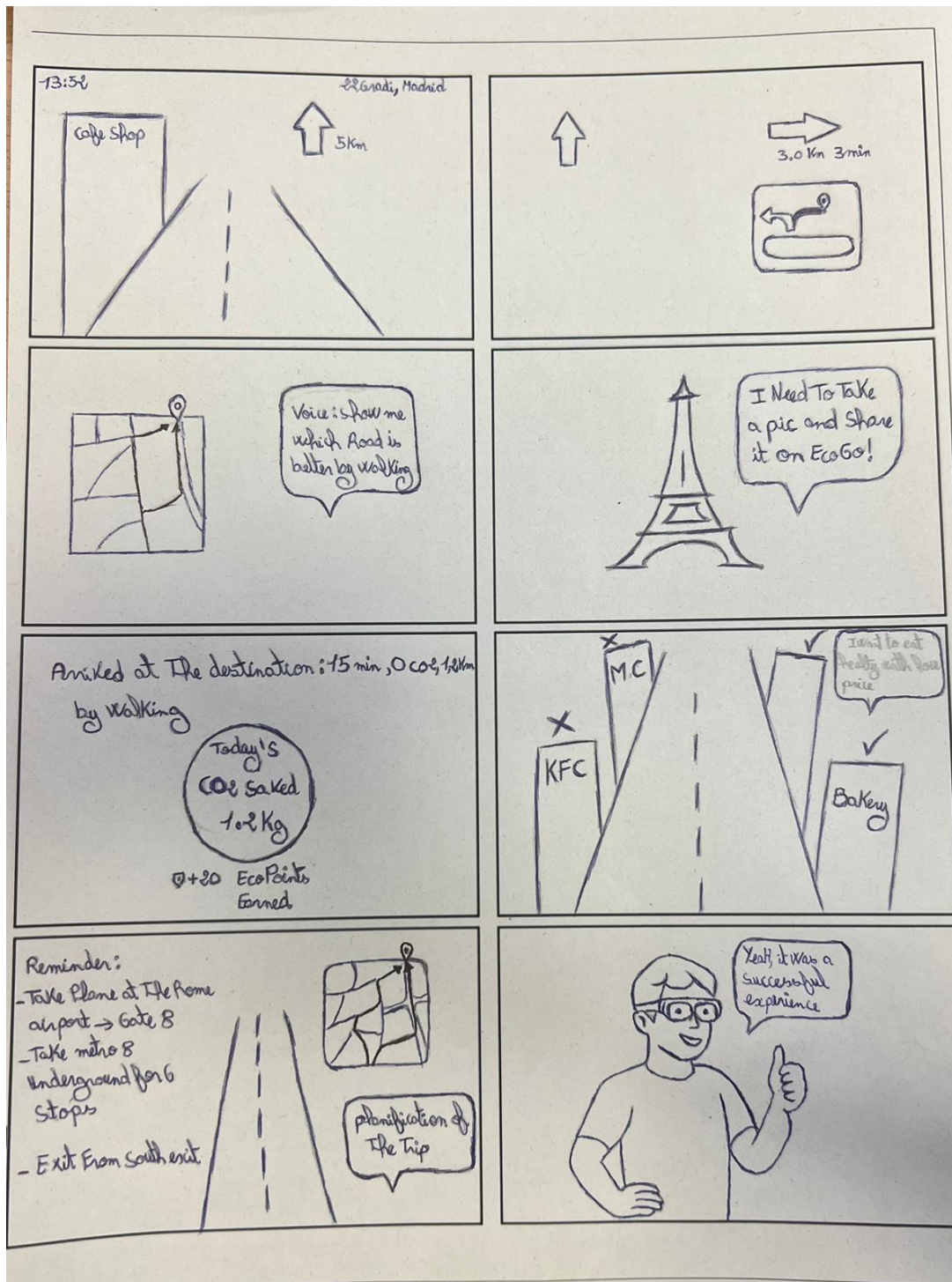
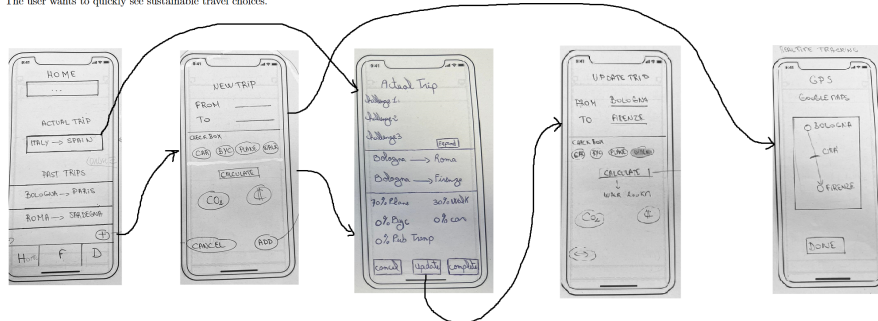


Figure 2: Real example illustrating the ecoGo user journey with smart glasses.

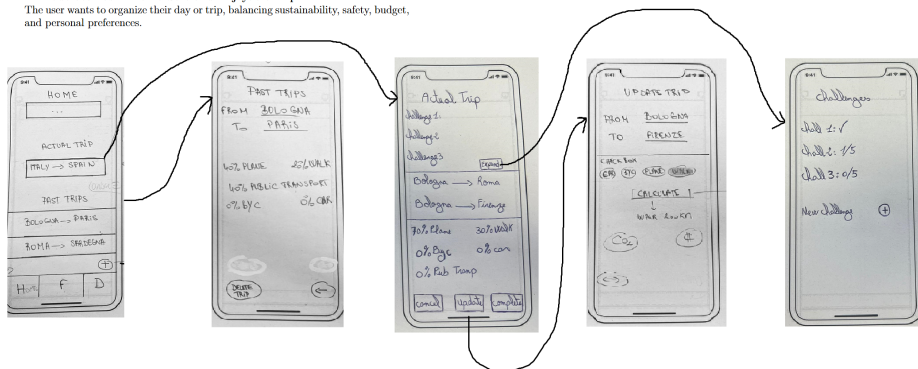
4 Paper prototypes 1

4.1 High-level flow of the “screens”

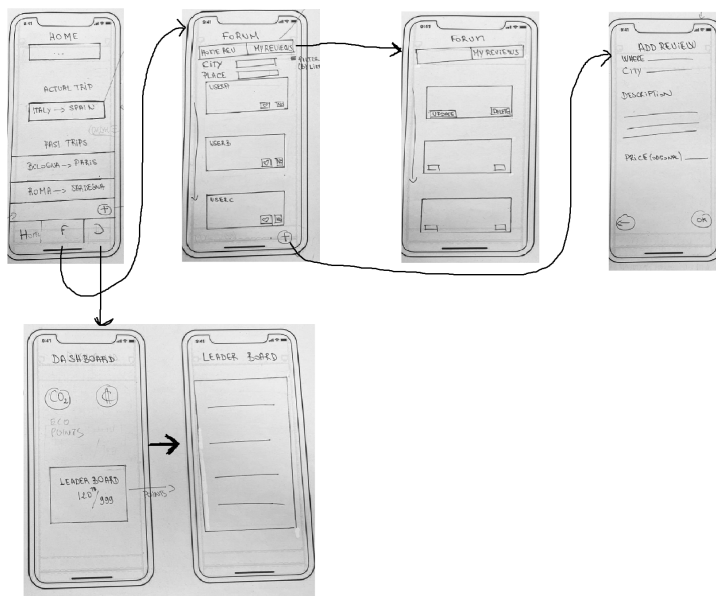
- **1. Simple Task: Discover Eco-Friendly Options**
The user wants to quickly see sustainable travel choices.



- **2. Moderate Task: Plan a Safe and Enjoyable Trip**
The user wants to organize their day or trip, balancing sustainability, safety, budget, and personal preferences.



- **3. Complex Task: Track, Reflect, and Share Sustainable Actions**
The user wants to monitor their sustainable choices, see their impact, complete optional challenges, and share experiences with others.



4.2 Connection with 3 tasks

1. Simple Task: Discover Eco-Friendly Options

The first storyboard sequence illustrates how the user quickly discovers sustainable travel options directly from the home screen. The interface shows recent trips, eco-friendly suggestions, and shortcuts to start a new planning session.

By selecting a trip, the user immediately accesses information such as route options, CO₂ saved, and recommended green choices.

2. Moderate Task: Plan a Safe and Enjoyable Trip

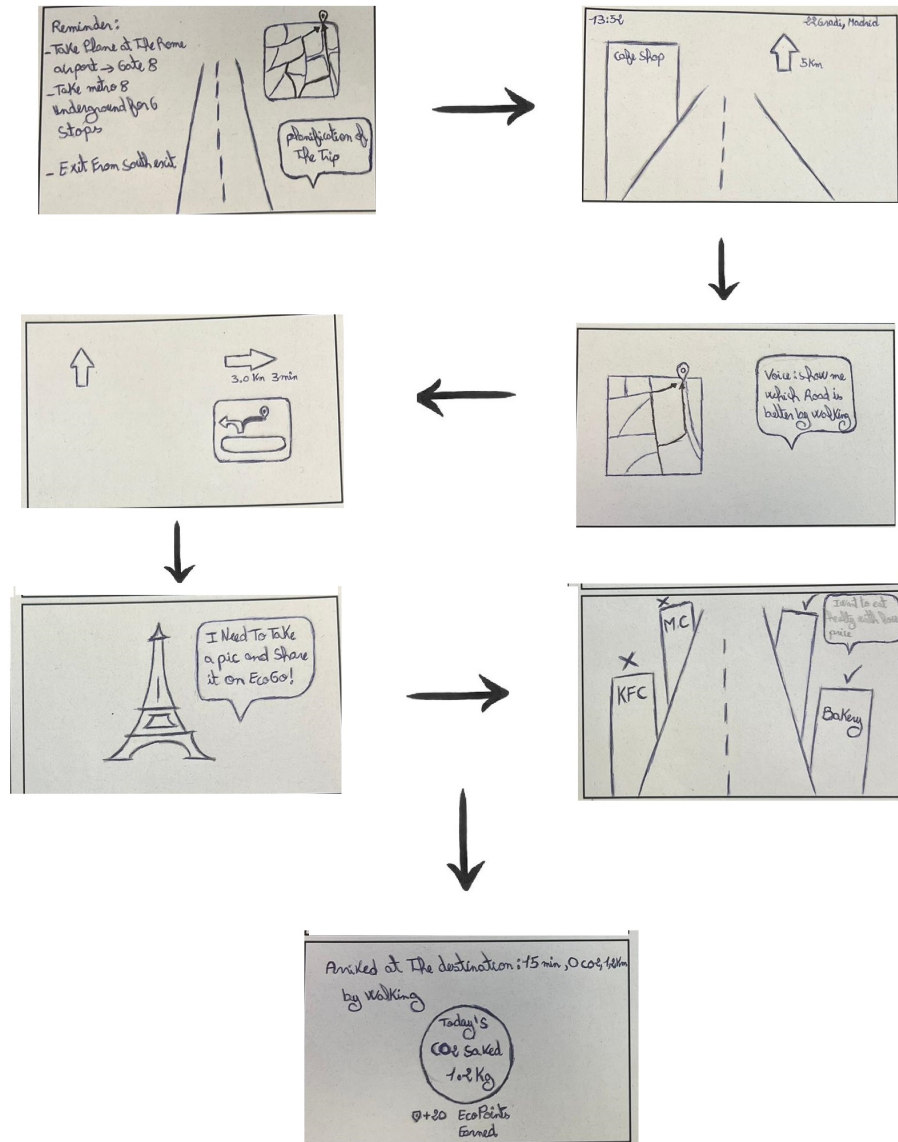
The second set of screens demonstrates how the user plans a trip by balancing sustainability, personal preferences, and safety. The app guides the user through selecting departure and arrival points, choosing transport modes, and reviewing estimated CO₂ impact.

3. Complex Task: Track, Reflect, and Share Sustainable Actions

The third storyboard focuses on self-reflection and motivation. These screens show leaderboards and challenges, giving the user opportunities to compare progress, stay motivated, and share achievements with others. The flow highlights how the design supports long-term engagement with sustainability by making the user's impact visible and meaningful.

5 Paper prototypes 2

5.1 High-level flow of the “screens”



5.2 Connection with 3 tasks

First, it supports the **Simple Task: Discover Eco-Friendly Options** by displaying nearby sustainable places during navigation. As the user walks, the glasses highlight eco-friendly cafés, shops, or points of interest, allowing them to effortlessly notice green alternatives without needing to check a phone or map.

Second, the prototype reflects the **Moderate Task: Plan a Safe and Enjoyable Trip**. The glasses provide turn-by-turn guidance, reminders, and contextual information that help the user follow their planned route safely. The system warns about distance, direction, and estimated time, while also showing options that align with sustainability and

personal preferences.

Finally, the smart-glasses concept embodies the **Complex Task: Track, Reflect, and Share Sustainable Actions**. At the end of the journey, the system summarizes the CO₂ saved and awards eco-points, giving immediate feedback on the user's positive environmental impact. This supports reflection and progress tracking, and the eco-points could later be shared or used to motivate long-term sustainable behavior. The last frame of the storyboard directly illustrates this reflective moment.

Real example: Amazon recently unveiled their own smart glasses, currently being tested by delivery drivers. These wearables include built-in navigation features and the ability to scan packages. For reference, see the video here: https://fb.watch/DLH_VLafCo/